

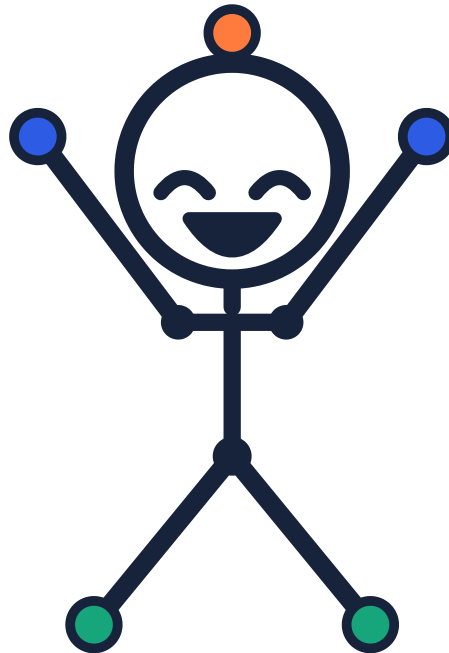


Young Innovators
FOR CHANGE
innovateyouth.org

A FULL-YEAR ENTERPRISE PROGRAM · GRADE 5

The Keeping Year

36 weekly assignments. What a thing costs to keep, what it costs each time you use it, and what happens when somebody else pays half.



This book belongs to

Something I share, with

A thing I own, bought on

Teacher

School year

Free to print, copy and use. No license, no account, no cost to any student or school.

The price is not the cost.

Two pages a week, front and back. About thirty-five minutes.

The shape of a week

Page 1 teaches one idea and gives you something to do with it. **Page 2** is practice, a question for home, and a harder challenge. No computer needed.

Weeks 1 to 12 put the price tag in its place: what a thing costs to keep, what it costs each time you use it, and how to compare two options over the same stretch of time.

Weeks 13 to 24 are money with other people, which is where it usually goes wrong.

Weeks 25 to 36 are about deciding well when a decision has more than one number in it.

The boxes on the page

IN CLASS

The Grade 5 standard your class is most likely on that week. This book sits beside your math lesson, not against it.

MONEY WARM-UP

Two minutes of arithmetic with money in it. Decimals multiplied and divided, and fractions of an amount, which is what 5.NBT.7 and 5.NF are for.

WORDS TO KEEP

One or two words pinned down properly. Adults use these loosely; you will not have to.

COST BOOK

The one thing you build all year. 15 pages by week 36, every one of them about your own things and your own agreements.

FROM YOUR BOOK

Which earlier pages this week needs. Week 33 is impossible without week 32.

TALK ABOUT IT

Two questions for someone at home. Five minutes, and the part that makes the week stick.

STUCK? TRY THIS

A way in when the page stops making sense.

CHALLENGE ZONE

Harder, and usually open. For whoever finishes early.

Three things this book does not do

Nothing is sold. No product, no customer, no market day. That is The Numbers Year, also Grade 5 and a different book.

Nobody is asked what anything costs at their house. Every amount is the student's own or is printed on the page.

No decision is marked right or wrong. What can be marked is whether the arithmetic behind it is.

WEEK	TITLE	STRAND	WHAT IT SITS BESIDE IN CLASS
TRIMESTER 1 · WEEKS 1–12 · THE PRICE IS NOT THE COST			
1	The Price Is Not the Cost	Running Costs	Operations with decimals to hundredths (5.NBT.7)
2	What It Costs to Keep	Running Costs	Operations with decimals to hundredths (5.NBT.7)
3	Price Plus Keeping	Running Costs	Multiply and divide multi-digit numbers (5.NBT.5, 5.NBT.7)
4	The Cheap One That Costs More	Running Costs	Operations with decimals to hundredths (5.NBT.7)
5	Cost Per Use	Per Use	Divide with decimals (5.NBT.7)
6	The Thing You Used Twice	Per Use	Divide with decimals (5.NBT.7)
7	How Long Will It Last?	Per Use	Divide with decimals (5.NBT.7)
8	Two Options, One Period	Comparing	Generate and compare two patterns (5.OA.3)
9	Where They Cross	Comparing	Patterns and relationships (5.OA.3)
10	The Number That Is Not Money	Comparing	Convert measurement units (5.MD.1)
11	Cheap, Good, or Soon	Comparing	Operations with decimals to hundredths (5.NBT.7)
12	The Cost Detective	Review	Operations with decimals to hundredths (5.NBT.7)
TRIMESTER 2 · WEEKS 13–24 · MONEY WITH OTHER PEOPLE			
13	Buying It Together	Together	Divide with decimals (5.NBT.7)
14	Not Everybody Pays the Same	Together	Add fractions with unlike denominators (5.NF.1)
15	What Each Share Comes To	Together	Multiply a fraction by a whole number (5.NF.4, 5.NF.6)
16	Share by Use	Together	Multiply a fraction by a whole number (5.NF.6)
17	Whose Is It?	Together	A fraction as division (5.NF.3)
18	You Owe, or You Are Owed	Owing	Operations with decimals to hundredths (5.NBT.7)

WEEK	TITLE	STRAND	WHAT IT SITS BESIDE IN CLASS
19	Paying It Back in Pieces	Owing	Divide with decimals (5.NBT.7)
20	The Friend Who Forgot	Owing	Operations with decimals to hundredths (5.NBT.7)
21	Lending, and Whether To	Owing	A fraction as division (5.NF.3)
22	An Agreement, in Writing	Together	Operations with decimals to hundredths (5.NBT.7)
23	When It Breaks	Together	Multiply a fraction by a whole number (5.NF.6)
24	The Sharing Page	Review	Word problems with unlike denominators (5.NF.2)
TRIMESTER 3 · WEEKS 25–36 · DECIDING WELL			
25	The Decision With Two Numbers	Deciding	Operations with decimals to hundredths (5.NBT.7)
26	And Then a Third	Deciding	Operations with decimals to hundredths (5.NBT.7)
27	What Would Change Your Mind?	Deciding	Operations with decimals to hundredths (5.NBT.7)
28	Deciding for Later You	Deciding	Operations with decimals to hundredths (5.NBT.7)
29	Every Month, Forever	Per Use	Multiply and divide multi-digit numbers (5.NBT.5, 5.NBT.7)
30	Own It or Borrow It	Per Use	Divide with decimals (5.NBT.7)
31	The Real Comparison	Comparing	Operations with decimals to hundredths (5.NBT.7)
32	A Decision You Already Made	Deciding	Operations with decimals to hundredths (5.NBT.7)
33	Was It Worth It?	Deciding	Operations with decimals to hundredths (5.NBT.7)
34	The One You Would Do Again	Deciding	Operations with decimals to hundredths (5.NBT.7)
35	Your Cost Book, Finished	Review	Operations with decimals to hundredths (5.NBT.7)
36	The Price Is Not the Cost	Review	Operations with decimals to hundredths (5.NBT.7)

The Price Is Not the Cost

What you pay at the counter is the first number, not the only one, and usually not the biggest.

IN CLASS THIS WEEK

Operations with decimals to hundredths (5.NBT.7)

MONEY WARM-UP • 2 MINUTES

A \$40 thing costs \$5 a month to keep. What is a year? _____

WORDS TO KEEP

running cost

What a thing keeps costing after you have paid for it.

the whole cost

The price plus everything it costs to keep, over the time you will have it.



Look at the whole column

One costs \$40 and \$5 a month. One costs \$90 and \$1 a month. Over a year that is **\$100** and **\$102**, which is almost the same money.

Almost nobody compares those two things that way. They compare \$40 with \$90 and feel like they saved fifty dollars, because the price is the number written on the tag and the running cost is not written anywhere.

This year the price is the bottom block. Everything the thing goes on costing gets stacked on top of it, and the number you compare is the height of the whole column.

Price, or whole cost?

	Price	Each month	12 months of keeping	The whole cost
a	\$40.00	\$5.00	\$60.00	\$100.00
b	\$90.00	\$1.00		
c	\$25.00	\$3.50		
d	\$120.00	\$0.00		
e	\$60.00	\$2.25		

Put the five in order by price. Now put them in order by whole cost. How many changed places?

► ADD TO YOUR COST BOOK

Start your Cost Book with this page: one thing you own, its price, and what it has cost you since.

TALK ABOUT IT

Find one thing at home that costs money after you bought it.

Why does the shop put the price on the tag and not the running cost?

**STUCK? TRY THIS**

Multiply the monthly amount by 12 first. Then add the price. Two steps, in that order.

★ CHALLENGE ZONE

Design two things where one has a price twice the other and a whole cost half the other. What has to be true about the running costs?

What It Costs to Keep

Naming the running costs, because a cost nobody has named does not get counted.

IN CLASS THIS WEEK

Operations with decimals to hundredths (5.NBT.7)

MONEY WARM-UP • 2 MINUTES

$\$2.40 + \$1.75 + \$0.90 = \underline{\hspace{2cm}}$

WORDS TO KEEP

running cost

What a thing keeps costing after you have paid for it.

the whole cost

The price plus everything it costs to keep, over the time you will have it.

RECEIPT	
charging it	\$2.40
a case	\$1.75
cleaning it	\$0.90

Four kinds, and people forget three

Things it needs to work. Things it needs to keep working. Things that get used up. And what it costs to replace when it stops.

A bike is the clearest case. It needs air in the tires, which is nearly free. It needs a chain oiled and a tube patched, which is not. It uses up brake pads. And in six years it is too small for you.

Nobody thinks about the fourth one when they are buying, which is why week 7 exists. If a thing has to be replaced, the replacement is part of what it costs to have one of these, and it always was.

What does it cost to keep?

	The thing	Needs to work	Gets used up	Wears out in
a	a bike	air	brake pads	about 6 years
b	a games console			
c	a pair of school shoes			
d	a pet fish			
e	a phone			

One of these five costs almost nothing to keep and one costs more to keep than it cost to buy. Which are they?

TALK ABOUT IT

Ask somebody at home what one thing costs to keep, not to buy.

Which running cost is easiest to forget about? Why that one?

**STUCK? TRY THIS**

Ask what happens if you never spend another dollar on it. Whatever stops working is the running cost.

★ CHALLENGE ZONE

A thing is free to buy and expensive to keep. Name three real examples and say who is paying for them.

Price Plus Keeping

One number for the whole thing, so two things can actually be compared.

IN CLASS THIS WEEK

Multiply and divide multi-digit numbers (5.NBT.5, 5.NBT.7)

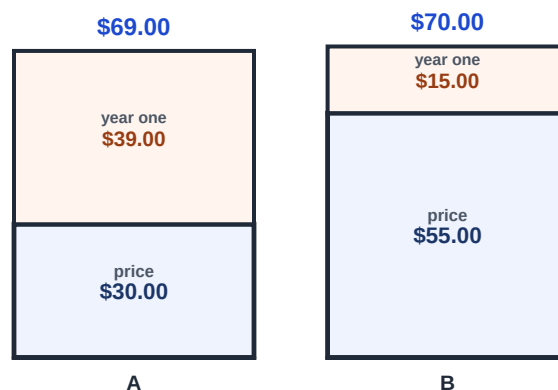
MONEY WARM-UP • 2 MINUTES

\$3.25 a month for 12 months is _____

WORDS TO KEEP

over a year

Twelve months of running cost, added to the price. One number you can compare.



Add it up once, properly

\$30.00 and \$3.25 a month. Twelve months is \$39.00, so the first year costs \$69.00 altogether.

The other one is \$55.00 and \$1.25 a month, which is \$15.00 a year, so \$70.00. A dollar apart, and their price tags are twenty-five dollars apart.

Notice what the whole cost depends on: **how long you keep it**. Over one year these are the same. Over three years they are not close, and week 8 is about choosing the stretch of time before you do the arithmetic rather than after.

The first year

	Price	Each month	A year of keeping	First year, altogether
a	\$30.00	\$3.25	\$39.00	\$69.00
b	\$55.00	\$1.25		
c	\$18.00	\$5.75		
d	\$140.00	\$0.75		
e	\$72.00	\$2.00		

Two of these come within five dollars of each other over the first year and their prices are twenty-five dollars apart. Which two?

► ADD TO YOUR COST BOOK

Add a page headed Price Plus Keeping, with two real things costed over their first year and the two columns side by side.

TALK ABOUT IT

Work out the first-year cost of something at home.

Is a year the right length to compare over? What would you use instead?

**STUCK? TRY THIS**

Twelve times the monthly amount, then add the price. Keep the cents in all the way to the end.

★ CHALLENGE ZONE

Two things cost the same over one year. Over three years one costs twice the other. Invent them.

The Cheap One That Costs More

The reversal, and it happens more often than anybody expects.

IN CLASS THIS WEEK

Operations with decimals to hundredths (5.NBT.7)

MONEY WARM-UP · 2 MINUTES

$\$18 + 12 \times \$6.00 = \underline{\hspace{2cm}}$

WORDS TO KEEP

over a year

Twelve months of running cost, added to the price. One number you can compare.



Eighteen dollars, and then seventy-two

The \$18.00 one costs \$6.00 a month, so a year is \$72.00 and the whole thing is \$90.00. The \$72.00 one costs 50 cents a month, so a year is \$6.00 and the whole thing is \$78.00.

The cheap one costs twelve dollars more. It was never the cheap one; it was the one with the cheap price, and those are different things.

This is not a rule that expensive things are secretly cheaper. Sometimes the expensive one is just expensive. The rule is that you cannot know which case you are in without the arithmetic, and the arithmetic takes about a minute.

Which one really costs more?

	Cheap price	Its monthly	Dear price	Its monthly	Which costs more over a year?
a	\$18.00	\$6.00	\$72.00	\$0.50	the cheap one
b	\$25.00	\$2.00	\$40.00	\$1.00	
c	\$60.00	\$1.50	\$85.00	\$0.25	
d	\$12.00	\$8.00	\$95.00	\$0.00	
e	\$30.00	\$3.00	\$50.00	\$2.00	

In three rows the cheaper price is also the cheaper year. In two it is not. Which two, and what do those two have in common?

 TALK ABOUT IT

Tell somebody at home about a cheap thing that turned out expensive.

Does knowing this change what you would buy, or just how you would feel about it?

**STUCK? TRY THIS**

Work out both whole years first. Compare only at the end, and never compare the prices on their own.

★ CHALLENGE ZONE

A thing has a cheap price and a cheap running cost and is still the worse buy. How is that possible? Week 7 has the answer if you want to look ahead.

Cost Per Use

The whole cost, divided by the number of times you actually used it.

IN CLASS THIS WEEK

Divide with decimals (5.NBT.7)

MONEY WARM-UP · 2 MINUTES

\$60.00 divided by 200 uses is _____

WORDS TO KEEP

cost per use

The whole cost divided by the number of times you used it.



The number that settles an argument

\$60.00 shoes worn 200 times cost **30 cents** a wearing. \$25.00 shoes worn 20 times cost **\$1.25** a wearing.

The expensive shoes are four times cheaper. Not cheaper to buy, cheaper to use, and using them was the point of having them.

This is the number to reach for when somebody says a thing is too expensive, and it is also the number that will tell you honestly about something in your own room that you wanted very badly and have used twice. It cuts both ways, which is why it is worth knowing.

Work out the cost per use

	Whole cost	Times used	Cost per use
a	\$60.00	200	\$0.30
b	\$25.00	20	
c	\$90.00	45	
d	\$14.00	7	
e	\$120.00	500	

Put the five in order by whole cost. Now by cost per use. The most expensive thing on this page is the cheapest to use. Which is it?

► ADD TO YOUR COST BOOK

Add a page headed Cost Per Use, with three things you own, honestly counted, and the cheapest one circled.

TALK ABOUT IT

Work out the cost per use of something you own and use a lot.

Is there anything you own where you would rather not do this?

**STUCK? TRY THIS**

Divide the whole cost by the number of uses. The answer is money, so it goes to the cent.

★ CHALLENGE ZONE

Two things have the same cost per use and one cost ten times as much. What is true about how often each gets used?

The Thing You Used Twice

The honest version of the same arithmetic, done on something that did not work out.

IN CLASS THIS WEEK

Divide with decimals (5.NBT.7)

MONEY WARM-UP · 2 MINUTES

\$48.00 divided by 2 uses is _____

WORDS TO KEEP

cost per use

The whole cost divided by the number of times you used it.



Twenty-four dollars a go

Something that cost \$48.00 and got used twice cost \$24.00 each time. That is not a small number and it does not feel like one either.

Everybody has one of these. It is not a moral failure and this page is not a telling off. It is a measurement, and the reason to take it is that it is the only honest input to the next decision.

Here is the useful part. Before you buy something, guess how many times you will use it, then halve your guess. People are reliably wrong in the same direction, and halving is a rough correction for a real bias rather than pessimism.

The uncomfortable table

	What it cost	Times you guessed	Times you really used it	Cost per use, really
a	\$48.00	20	2	\$24.00
b	\$30.00	50	5	
c	\$16.00	10	8	
d	\$75.00	100	12	
e	\$9.00	4	1	

Work out what each would have cost per use if the guess had been right. In which row does the guess matter least?

**TALK ABOUT IT**

Find one thing you use much less than you expected to. Just find it.

Should you get rid of a thing you barely use, or does that make it worse?

**STUCK? TRY THIS**

Divide by the real number, not the hoped-for one. Then do it again with the guess and compare.

★ CHALLENGE ZONE

You are about to buy something for \$40 and you guess 30 uses. Halve the guess. What is the cost per use now, and does it change your answer?

How Long Will It Last?

Replacing it is part of what it costs to have one, and it always was.

IN CLASS THIS WEEK

Divide with decimals (5.NBT.7)

MONEY WARM-UP · 2 MINUTES

\$84.00 over 3 years is _____ a year

WORDS TO KEEP

how long it lasts The number of uses or years before it has to be replaced. A guess, written down.



Three cheap ones is not cheaper than one good one

A \$30.00 thing that lasts a year costs \$90.00 over three years. A \$72.00 thing that lasts three years costs \$72.00 over three years.

Same three years, eighteen dollars apart, and the cheap one is the expensive one again. This is the fourth running cost from week 2, the one nobody names, and it is usually the largest.

How long a thing lasts is a guess. Write the guess down anyway. A written guess can be wrong in a way you can learn from; an unwritten one just quietly becomes whatever you wanted it to be.

Cost per year

	Price	Lasts	Cost per year	Over 6 years
a	\$30.00	1 year	\$30.00	\$180.00
b	\$72.00	3 years		
c	\$84.00	2 years		
d	\$150.00	6 years		
e	\$45.00	18 months		

Over six years, which is cheapest? Is it the one with the cheapest price, or the one with the longest life, or neither?

► ADD TO YOUR COST BOOK

Add a page headed How Long It Lasts, with three things you own and your written guess for each.

TALK ABOUT IT

Ask somebody at home about something they bought that lasted a long time.

Can you tell how long a thing will last before you buy it? What would you look at?

**STUCK? TRY THIS**

Divide the price by the number of years to get the cost per year. Then multiply by six so every row covers the same time.

★ CHALLENGE ZONE

A \$30 one lasts a year and a \$150 one lasts six. They cost the same over six years. Which would you buy, and what does your answer depend on?

Two Options, One Period

A comparison over two different stretches of time is not a comparison at all.

IN CLASS THIS WEEK

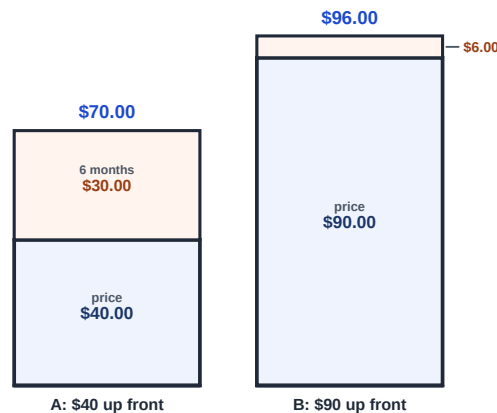
Generate and compare two patterns (5.OA.3)

MONEY WARM-UP · 2 MINUTES

$\$40 + 6 \times \5 against $\$90 + 6 \times \1 . Which is less? _____

WORDS TO KEEP

the same period Two options can only be compared over the same stretch of time. Different stretches are not a comparison.



Pick the stretch first

Over six months, A costs \$70.00 and B costs \$96.00. Over two years, A costs \$160.00 and B costs \$114.00.

Both of those are true and they point opposite ways. Neither is a trick. The answer to which is cheaper depends entirely on how long you are going to have it, and that is a question about you rather than about the two things.

So decide the stretch of time **before** you do any arithmetic. Choosing it afterwards, once you can see which answer you like, is how people talk themselves into things and it does not feel dishonest while it is happening.

The same options over four different stretches

	How long	Option A: \$40 + \$5 a month	Option B: \$90 + \$1 a month	Cheaper
a	6 months	\$70.00	\$96.00	A
b	1 year			
c	2 years			
d	3 years			
e	5 years			

A is cheaper in some rows and B in others. Between which two rows does the answer change?

 TALK ABOUT IT

Ask somebody at home how long they expected something to last when they bought it. Is it cheating to pick the time period that makes your answer come out right?

**STUCK? TRY THIS**

Work out one option all the way down the column, then the other. Comparing row by row invites mistakes.

★ CHALLENGE ZONE

Write a rule for choosing the stretch of time that does not depend on which answer you are hoping for.

Where They Cross

The month the cheaper option stops being the cheaper one, found rather than guessed.

IN CLASS THIS WEEK

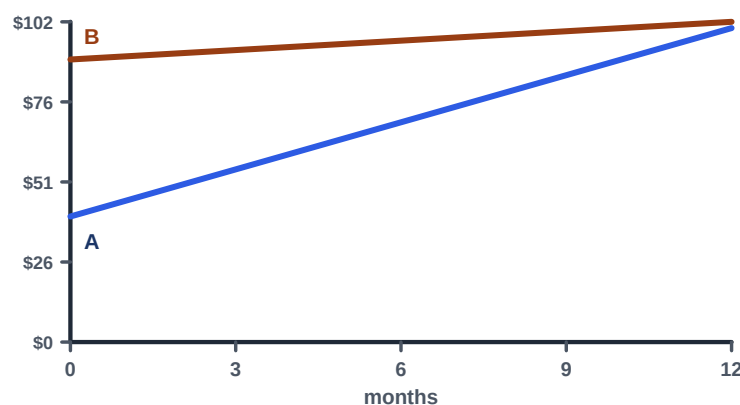
Patterns and relationships (5.OA.3)

MONEY WARM-UP • 2 MINUTES

\$40 + \$5m and \$90 + \$1m. At what m are they equal? _____

WORDS TO KEEP

where they cross The point where the cheaper option stops being the cheaper one.



Two lines, one crossing

A starts lower and climbs faster. B starts higher and barely climbs. Somewhere they meet, and after that A is the expensive one forever.

You can find it by building the table month by month until the numbers swap, which always works and is worth doing once. A is \$40 ahead at the start and closes \$4 of that every month, so it takes **12 and a half months**, and at 13 months B has won.

The crossing is the only number in the comparison that does not depend on your opinion. Once you know it, the decision becomes a single honest question: will I still have this in thirteen months?

Find the crossing

	Option A	Option B	Gap at the start	Closed each month	They cross at
a	\$40 + \$5	\$90 + \$1	\$50	\$4	12.5 months
b	\$20 + \$6	\$56 + \$2			
c	\$30 + \$4	\$66 + \$1			
d	\$50 + \$3	\$80 + \$2			
e	\$15 + \$9	\$99 + \$2			

Four of these cross at a whole number of months. One does not. Which, and what would you write for it?

► ADD TO YOUR COST BOOK

Add a page headed Where They Cross, with two real options, both lines drawn, and the crossing month marked.

TALK ABOUT IT

Explain the crossing point to somebody at home using one of these rows.

If you are not sure how long you will keep something, which side of the crossing is safer to be wrong on?

**STUCK? TRY THIS**

Take the gap between the two prices. Divide it by the difference between the two monthly amounts. That is how many months to close it.

★ CHALLENGE ZONE

Both options cost the same every month. Where do they cross? Say what that means about the decision.

The Number That Is Not Money

Some reasons have no dollar amount. They still go in the decision.

IN CLASS THIS WEEK

Convert measurement units (5.MD.1)

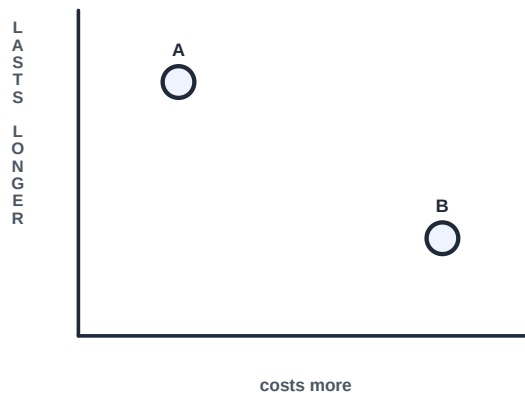
MONEY WARM-UP · 2 MINUTES

A costs \$70 and takes 20 minutes. B costs \$95. Cheaper? _____

WORDS TO KEEP

not money

A reason that matters and has no dollar amount. It still goes in the decision.



Do not pretend it has a price

A costs \$70.00 and takes twenty minutes of your time every week. B costs \$95.00 and takes none.

There is a way of answering this that puts a price on your time, decides an hour is worth a certain number of dollars, and turns the whole thing into one number. Adults do it. It is a real method and it is not the one this book teaches, because the price it puts on your time is invented and then the invented number quietly decides the answer.

Do this instead. Write the money. Beside it, write the thing that has no price, in words. Then decide, knowing you decided rather than having the arithmetic decide for you.

Money, and the other thing

	Option A	Option B	Cheaper	What the cheaper one costs you that is not money
a	\$70, 20 min a week	\$95, no time	A	about 17 hours a year
b	\$40, breaks often	\$65, reliable		
c	\$12, lasts a year	\$30, lasts five		
d	\$25, borrowed	\$50, yours		
e	\$80, heavy	\$110, light		

In one of these rows the thing that is not money is worth more than the money to almost everybody. Which row, and why that one?

TALK ABOUT IT

Ask somebody at home about a time they paid more on purpose. What for?

Is your time worth money? Does the answer change on a Saturday?

**STUCK? TRY THIS**

Work out the money first and get it out of the way. Then write the other reason in a sentence, not as a number.

★ CHALLENGE ZONE

Somebody says your time is worth \$10 an hour and uses it to settle row a. What is right about that and what is wrong with it?

Cheap, Good, or Soon

You can usually have two. The useful skill is saying out loud which one you are giving up.

IN CLASS THIS WEEK

Operations with decimals to hundredths (5.NBT.7)

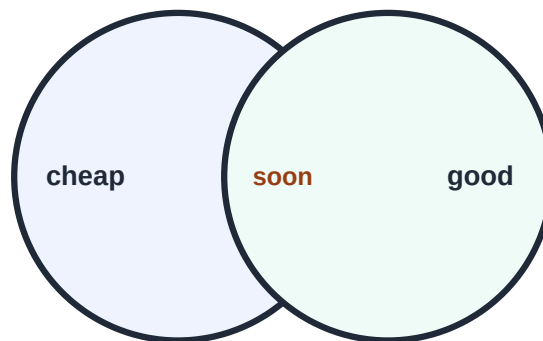
MONEY WARM-UP • 2 MINUTES

\$18 now, \$12 in three weeks. What does waiting save? _____

WORDS TO KEEP

not money

A reason that matters and has no dollar amount. It still goes in the decision.



Name the one you dropped

Cheap, good, and soon. Most real choices let you have two of them, and the third one is what you are paying with.

Waiting three weeks to save six dollars means giving up **soon**. Buying the flimsy one today means giving up **good**. Buying the good one today means giving up **cheap**. None of those is wrong.

The mistake is not picking badly, it is not noticing you picked. Somebody who says the good one today was too expensive has usually decided that soon mattered most, without ever saying so, and will be annoyed about the money for weeks.

Which one did you give up?

	The choice	Cheap?	Good?	Soon?	Which one you gave up
a	wait 3 weeks and save \$6	yes	yes	no	soon
b	buy the flimsy one today				
c	buy the good one today				
d	borrow a friend's for now				
e	make one yourself				

Row d gets all three and is still not always the right answer. What is it costing that is not on this list?

TALK ABOUT IT

Tell somebody at home about a choice where you gave up soon. Was it worth it?
Which of the three do you give up most often? Is that a habit or a decision?

**STUCK? TRY THIS**

For each row, mark the three columns yes or no first. The last column is whichever one you wrote no in.

★ CHALLENGE ZONE

Find a real choice where you can have all three. What is unusual about it?

The Cost Detective

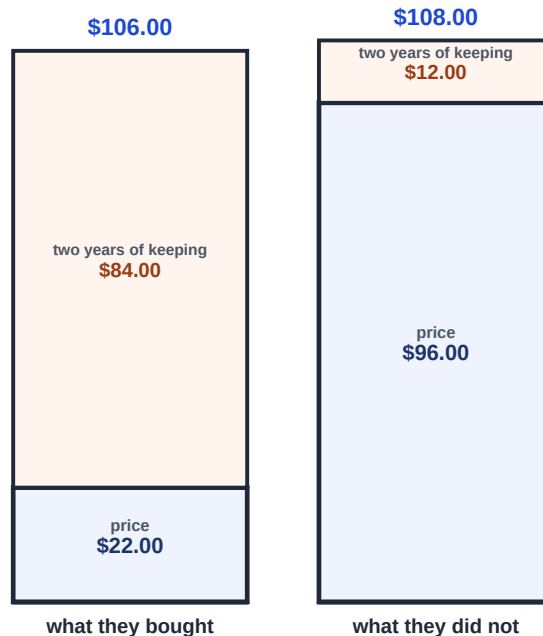
Everything from weeks 1 to 11, on one decision somebody else got wrong.

IN CLASS THIS WEEK

Operations with decimals to hundredths (5.NBT.7)

MONEY WARM-UP · 2 MINUTES

$$\$22 + 24 \times \$3.50 = \underline{\hspace{2cm}}$$



Four things they did not work out

Somebody chose the \$22.00 one over the \$96.00 one because it was seventy-four dollars cheaper. It costs \$3.50 a month to keep; the other costs 50 cents. They kept it for two years and used it about forty times.

Work out what each would have cost over the two years they actually had it. Then work out the cost per use of the one they bought. Then say what the crossing month was, and whether they passed it.

Write your four findings in sentences. A detective says what happened, not just which number is bigger.

The evidence

	The question	Your answer
a	What did the one they bought cost over two years?	
b	What would the other one have cost over the same two years?	
c	At forty uses, what did the one they bought cost per use?	
d	In which month did the two options cross?	
e	Did they keep it past the crossing month?	

Now write the one sentence you would have said to them in the shop. It has to contain a number.

**TALK ABOUT IT**

Take your sentence home and try it on somebody. Does it land?

Would knowing all this have changed their mind, or just made them feel worse?

**STUCK? TRY THIS**

Do part a and part b before anything else. Everything on this page is easier once both whole costs are written down.

★ CHALLENGE ZONE

Change one number on this page so that the person turns out to have chosen correctly. Which number did you pick, and why that one?

Buying It Together

One cost, more than one person, and the arithmetic is the easy part.

IN CLASS THIS WEEK

Divide with decimals (5.NBT.7)

MONEY WARM-UP · 2 MINUTES

\$45.00 split evenly between 3 people is _____

WORDS TO KEEP

split

One cost divided between the people paying it.



Even is a choice, not a default

\$45.00 between three people is \$15.00 each. That is the whole calculation and it takes four seconds.

Everything hard about this week happens before and after. Before: does everybody agree the thing is worth \$45.00, or has one person decided and the others are going along with it? After: what happens when one of them cannot pay this week.

Splitting evenly is fair when everybody wants it equally and will use it equally. That is often true and it is not automatic, which is what weeks 14 to 16 are about.

Split it evenly

	The cost	People	Each pays	Anything left over
a	\$45.00	3	\$15.00	nothing
b	\$30.00	4		
c	\$50.00	3		
d	\$28.00	8		
e	\$100.00	6		

Three rows do not divide evenly into cents. For each one, say who pays the extra penny and how you would decide that.

► ADD TO YOUR COST BOOK

Add a page headed Buying It Together, with one real shared cost, the people, and what each paid.

TALK ABOUT IT

Ask somebody at home about a time a group bought something together.

Who should decide what a shared thing is worth: the person paying most, or everybody equally?

**STUCK? TRY THIS**

Divide the cost by the number of people. If it does not come out to whole cents, the leftover is real and somebody has to carry it.

★ CHALLENGE ZONE

Four people split \$30.00 and one of them can only pay \$5. Write two different fair ways to finish the split, and say what each one assumes.

Not Everybody Pays the Same

An uneven split, written as fractions that still have to reach one whole.

IN CLASS THIS WEEK

Add fractions with unlike denominators (5.NF.1)

MONEY WARM-UP · 2 MINUTES

$$1/2 + 1/3 + 1/6 = \underline{\hspace{2cm}}$$

WORDS TO KEEP

an uneven share A split where the parts are not the same size, written as fractions that still add to one whole.



Unlike denominators, one whole

A half, a third and a sixth. Over six they are 3/6, 2/6 and 1/6, which add to 6/6. One whole, so the split is complete.

On \$36.00 that is \$18.00, \$12.00 and \$6.00. Notice that you can check the money the other way: the three amounts add to \$36.00, and if they did not, one of the fractions was wrong.

An uneven split needs a reason, and the reason is the whole point. Ana pays half because she wanted it most, or uses it most, or has the most money. Those are three completely different agreements and week 22 is about writing down which one you made.

Do the shares reach one whole?

	The shares	Over a common bottom	Add to	One whole?
a	$\frac{1}{2}, \frac{1}{3}, \frac{1}{6}$	$\frac{3}{6}, \frac{2}{6}, \frac{1}{6}$	$\frac{6}{6}$	yes
b	$\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$			
c	$\frac{2}{5}, \frac{2}{5}, \frac{1}{10}$			
d	$\frac{1}{3}, \frac{1}{3}, \frac{1}{4}$			
e	$\frac{3}{8}, \frac{3}{8}, \frac{1}{4}$			

One row does not reach one whole. How short is it, and what would happen in real life if three people paid those shares?

 TALK ABOUT IT

Work out your own share of something at home as a fraction.

Is an uneven split unfair? What would make it fair or unfair?

**STUCK? TRY THIS**

Find a bottom number all three fractions fit into. Rewrite them over it, then add the tops.

★ CHALLENGE ZONE

Three people split a cost as $\frac{1}{2}$, $\frac{1}{3}$ and something. What is the third share? Now do it for $\frac{1}{2}$, $\frac{1}{5}$ and something.

What Each Share Comes To

Turning the fractions into money, which is where a disagreement gets settled.

IN CLASS THIS WEEK

Multiply a fraction by a whole number (5.NF.4, 5.NF.6)

MONEY WARM-UP · 2 MINUTES

$\frac{1}{3}$ of \$36.00 is _____

WORDS TO KEEP

an uneven share A split where the parts are not the same size, written as fractions that still add to one whole.



Money does not have opinions

Half of \$36.00 is \$18.00. A third is \$12.00. A sixth is \$6.00. They add to \$36.00, so the split works.

Two people can argue about whether a third is a fair share for a long time. They cannot argue about whether a third of \$36.00 is \$12.00, and getting from the first argument to the second one is most of what this week does.

Always add the amounts up at the end. If they do not come to the cost, something is wrong with the fractions, and finding that out now is much better than finding it out at the register with three people watching.

What does each person pay?

	The cost	Ana	Ben	Cal	Do the three add to the cost?
a	\$36.00	\$18.00	\$12.00	\$6.00	yes
b	\$60.00				
c	\$24.00				
d	\$45.00				
e	\$50.00				

Ana always pays $\frac{1}{2}$, Ben $\frac{1}{3}$ and Cal $\frac{1}{6}$. In two rows somebody's share is not a whole number of cents. Which rows, and what would you do?

► ADD TO YOUR COST BOOK

Add a page headed Our Shares, with the fractions, the amounts, and the check that they add to the cost.

TALK ABOUT IT

Work out three shares of a real amount at home and check they add up.

If a share comes to \$8.33 and a third, who pays the extra cent?

**STUCK? TRY THIS**

Divide by the bottom number, then multiply by the top. Add all three at the end and compare with the cost.

★ CHALLENGE ZONE

Find a cost under \$100 where $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{6}$ all come out to whole dollars. What do all such costs have in common?

Share by Use

Splitting by how much each person uses it, which is a different kind of fair.

IN CLASS THIS WEEK

Multiply a fraction by a whole number (5.NF.6)

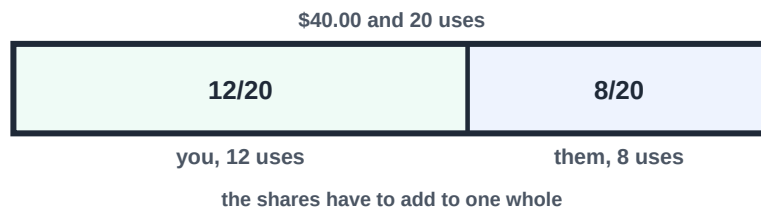
MONEY WARM-UP • 2 MINUTES

You used it 12 of 20 times. What fraction? _____

WORDS TO KEEP

share by use

Splitting by how much each person uses it rather than by how many people there are.



Count the uses, then split

Two people share a \$40.00 thing. One uses it 12 times, the other 8. That is $12/20$ and $8/20$, so \$24.00 and \$16.00.

Splitting by use is fairer than splitting by head count when the use is genuinely uneven, and it is only possible if somebody counted. Nobody counts unless it was agreed at the start, which is why this is a week about agreements as much as about fractions.

It also has a cost of its own. Counting every use of a shared thing between friends is a small ongoing unpleasantness, and sometimes an even split that is slightly wrong is worth more than an exact split that needs a tally on the wall.

Split by use

	The cost	You used	They used	Your share	Your money
a	\$40.00	12	8	12/20	\$24.00
b	\$30.00	5	5		
c	\$60.00	9	3		
d	\$25.00	1	4		
e	\$48.00	10	14		

In one row splitting by use gives the same answer as splitting evenly. Which, and what does that tell you about when the two methods differ?

 TALK ABOUT IT

Ask somebody at home whether they would want a shared thing split by use.

Would you rather pay slightly too much or count every time you used something?

**STUCK? TRY THIS**

Put your uses over the total uses. That fraction of the cost is your share.

★ CHALLENGE ZONE

You agreed to split by use and then used it far more than you expected. Are you worse off than if you had agreed to split evenly? Show it with numbers.

Whose Is It?

People paid different amounts. The thing is still one thing.

IN CLASS THIS WEEK

A fraction as division (5.NF.3)

MONEY WARM-UP • 2 MINUTES

You paid \$18 of \$30. What fraction of it is yours? _____

WORDS TO KEEP

share by use

Splitting by how much each person uses it rather than by how many people there are.



Owning three fifths of a thing

You paid \$18.00 of \$30.00, which is $\frac{3}{5}$. They paid $\frac{2}{5}$. So you own three fifths of it, and three fifths of a thing is not something you can put in your bag.

This is the part people skip and then argue about six months later. Who keeps it at night? What happens if one of you moves away? If it gets sold for \$10.00, who gets what? Those all have answers and the answers are much easier to agree on now than then.

Fractions can tell you what each person's share is worth. They cannot tell you whose shelf it lives on. That is an agreement, and week 22 is where you write one.

Who owns what

	Cost	You paid	Your fraction	If it sold for half, you get
a	\$30.00	\$18.00	$\frac{3}{5}$	\$9.00
b	\$40.00	\$10.00		
c	\$24.00	\$12.00		
d	\$50.00	\$45.00		
e	\$36.00	\$4.00		

In row d you paid nearly all of it. Should the other person get any say in what happens to it? Argue it both ways.

 TALK ABOUT IT

Ask somebody at home who owns a thing two people paid for.

Is owning most of something the same as it being yours?

**STUCK? TRY THIS**

Put what you paid over the whole cost and simplify. That fraction is your share of anything that happens to it later.

★ CHALLENGE ZONE

Two people pay $\frac{3}{5}$ and $\frac{2}{5}$. It breaks and is worth nothing. Who lost more? Now answer it again in dollars.

You Owe, or You Are Owed

The same fact from two sides, and both sides should be written in the same place.

IN CLASS THIS WEEK

Operations with decimals to hundredths (5.NBT.7)

MONEY WARM-UP · 2 MINUTES

You owe \$12.50 and are owed \$4.00. Net? _____

WORDS TO KEEP

owe

Money that is somebody else's and is currently with you.

owed

Money that is yours and is currently with somebody else.

Who	I owe them	They owe me	Left
Ana	\$12.50		
Ben		\$4.00	
Cal	\$3.00	\$3.00	

Write it down the same day

Money you owe is somebody else's money that happens to be with you. Money you are owed is your money that happens to be with somebody else.

Both go in the same ledger, in their own columns, on the day it happens. Not from memory on Sunday. Memory is generous about what you owe and sharp about what you are owed, and it is not doing that on purpose.

Look at Cal. Three dollars each way, so nothing is owed at all, and neither of you has to do anything. That row is the whole argument for keeping the ledger: without it, two people each remember being owed three dollars.

Fill in the ledger

	Who	I owe them	They owe me	So the position is
a	Ana	\$12.50	\$0.00	I owe \$12.50
b	Ben	\$0.00	\$4.00	
c	Cal	\$3.00	\$3.00	
d	Dee	\$6.75	\$2.25	
e	Eli	\$1.40	\$9.60	

Add the two money columns. Across all five people, are you a borrower or a lender, and by how much?

► ADD TO YOUR COST BOOK

Add a page headed Owed and Owing, with both columns, and the date beside every line.

TALK ABOUT IT

Write down anything you owe or are owed right now, both columns.

Is it rude to write down that a friend owes you money? What is the alternative?

**STUCK? TRY THIS**

Take the smaller from the bigger and say which way round it went. A row where they are equal comes to nothing, and nothing is an answer.

★ CHALLENGE ZONE

Three people each owe the next one \$5.00 in a circle. How much money needs to move? Now try it with four people.

Paying It Back in Pieces

A schedule you both agreed, and a box to cross off each time.

IN CLASS THIS WEEK

Divide with decimals (5.NBT.7)

MONEY WARM-UP · 2 MINUTES

\$21.00 paid back over 6 weeks is _____ a week

WORDS TO KEEP

in pieces

Paying back a little at a time, on a schedule you both agreed.



A schedule, not an intention

\$21.00 over six weeks is \$3.50 a week. Six boxes, and you cross one off every time you pay.

Saying you will pay somebody back when you can is not a plan and everybody involved knows it. An amount and a number of weeks is a plan, and it has the useful property that both people can see how it is going without either of them raising it.

Some amounts do not divide evenly. \$20.00 over six weeks is \$3.33 and a third, and a third of a cent does not exist. Pay \$3.34 for five weeks and \$3.30 at the end, or \$3.33 for five and \$3.35 at the end. Decide which, write it down, and the last week is not an argument.

Work out the schedule

	Owed	Weeks	Each week	The last week
a	\$21.00	6	\$3.50	\$3.50
b	\$20.00	6		
c	\$45.00	5		
d	\$14.00	4		
e	\$30.00	7		

Three of these do not divide evenly into cents. For each, write the amount you would pay in the last week so the total comes out exactly.

► ADD TO YOUR COST BOOK

Add a page headed Paying It Back, with the amount, the weekly payment, the number of weeks, and boxes to cross off.

💬 TALK ABOUT IT

Agree a schedule for something you owe, even if it is small.

Is it better to pay back a lot slowly or a little quickly?

**STUCK? TRY THIS**

Divide the amount by the weeks and round DOWN to the cent. Then the last week picks up whatever is left, and the total is exact.

★ CHALLENGE ZONE

You owe \$50.00 and can pay \$8.00 a week. How many weeks, and what is the last payment? Now the same for \$9.00 a week.

The Friend Who Forgot

What to do, and why writing it down was the kind thing rather than the suspicious one.

IN CLASS THIS WEEK

Operations with decimals to hundredths (5.NBT.7)

MONEY WARM-UP • 2 MINUTES

Owed \$8.00, paid back \$3.00 twice. Still owed? _____

WORDS TO KEEP

owe

Money that is somebody else's and is currently with you.

owed

Money that is yours and is currently with somebody else.

What happened	Amount	Still owed
you lent it	\$8.00	\$8.00
they paid you back, week 1	\$3.00	\$5.00
they paid you back, week 2	\$3.00	\$2.00

Almost nobody is lying

They remember paying you back. You remember them not. One of you is wrong and it is genuinely not obvious which.

People do not usually forget on purpose. Two payments and a half-remembered promise about a third is exactly the shape of thing a person loses track of honestly, and you would too.

Which is why the ledger is the friendly move rather than the distrustful one. Written down, this is arithmetic and takes ten seconds. Not written down, it is one person's memory against another's, and that is how eight dollars costs somebody a friend.

What is actually left?

	Lent	Paid back	Still owed	What you would say
a	\$8.00	\$3.00, \$3.00	\$2.00	
b	\$15.00	\$5.00		
c	\$6.50	\$6.50		
d	\$20.00	\$4.00, \$4.00, \$4.00		
e	\$9.00	nothing yet		

One row is settled and the person may not know it. What is the single most useful thing you could say to them?

 TALK ABOUT IT

Ask somebody at home whether they have ever lost money to a misunderstanding. How would you bring up money a friend owes you? Say the actual words.

**STUCK? TRY THIS**

Add up everything paid back and take it from what was lent. Do that before you decide what to say.

★ CHALLENGE ZONE

Write the sentence you would send a friend who owes you \$2.00 and thinks they owe nothing. Now write it again for \$50.00. What changed?

Lending, and Whether To

Decide before you hand it over, because afterwards you will decide differently.

IN CLASS THIS WEEK

A fraction as division (5.NF.3)

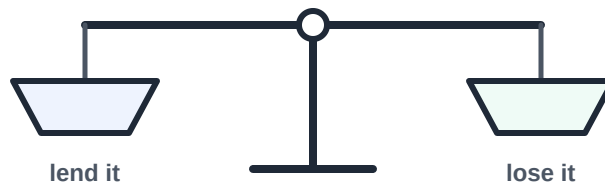
MONEY WARM-UP • 2 MINUTES

You lend \$10 and get back \$6. What did it cost you? _____

WORDS TO KEEP

lend

Hand over money you may not get back. Decide that before, not after.



Which is worth more to them?

The honest question

Would I be all right if this money never came back?

If the answer is yes, lend it and stop worrying, because you have already priced the bad outcome. If the answer is no, do not lend it, and you can say so without needing a better reason than that.

Notice what is not in that question. It does not ask whether your friend is trustworthy, because you will answer that one wrongly for people you like. It asks about you and about the amount, and both of those you can actually assess.

Most loans between friends are repaid. The ones that are not cost a friendship as well as the money, and that is the real price of getting this wrong.

Would you be all right?

	They ask for	You have	Would you be all right if it never came back?	Lend it?
a	\$2.00	\$20.00	yes	yes
b	\$18.00	\$20.00		
c	\$5.00	\$60.00		
d	\$40.00	\$45.00		
e	\$1.00	\$1.00		

Row e is a dollar and it is all the money you have. Row c is five dollars out of sixty. Which is the harder decision, and why is it not the bigger number?

 TALK ABOUT IT

Ask somebody at home for their rule about lending money to friends.

Is it different lending to family? Should it be?

**STUCK? TRY THIS**

Cover up who is asking. Look only at the amount and at what you have, and answer the question in the fourth column first.

★ CHALLENGE ZONE

Somebody offers to pay you back more than they borrowed, as a thank you. Should you take it? Say what you would do and why.

An Agreement, in Writing

What you both said, written down while you still agree about it.

IN CLASS THIS WEEK

Operations with decimals to hundredths (5.NBT.7)

MONEY WARM-UP · 2 MINUTES

Two people, \$50.00, split $\frac{3}{5}$ and $\frac{2}{5}$. Who pays what? _____

WORDS TO KEEP

an agreement

What you both said, written down while you still agree.



Five questions, before the money moves

Who pays what. Where does it live. Who can use it and when. What happens if it breaks. What happens if one of you wants out.

All five are easy to answer today and hard to answer the day you need them, because on that day one of you wants a particular answer. That is the whole reason to write it down now, and it is not about trust.

It does not need to be a document. Five lines in your Cost Book, read out loud to the other person, both of you saying yes. The reading out loud is the part that catches the thing you each assumed differently.

Write the five lines

	The question	What you both agreed
a	Who pays what?	
b	Where does it live?	
c	Who can use it, and when?	
d	What happens if it breaks?	
e	What happens if one of you wants out?	

Now read your five lines to the other person out loud. Write down anything they thought was different from what you wrote.

► ADD TO YOUR COST BOOK

Add a page headed Our Agreement, with all five lines and both names, dated.

TALK ABOUT IT

Read your agreement out loud to whoever you share the thing with.

Which of the five questions is the most awkward to ask? Why that one?

**STUCK? TRY THIS**

Answer question d before question c. Once you know who pays when it breaks, who gets to use it becomes much easier to agree.

★ CHALLENGE ZONE

Write question f, one this list does not cover. What does it protect against?

When It Breaks

The agreement either answers this or it does not, and now you find out which.

IN CLASS THIS WEEK

Multiply a fraction by a whole number (5.NF.6)

MONEY WARM-UP • 2 MINUTES

Repair is \$18.00, split $\frac{3}{5}$ and $\frac{2}{5}$. Your share? _____

WORDS TO KEEP

who pays

The question an agreement answers before anybody needs the answer.



Three answers, all of them defensible

Split the repair the way you split the purchase. Or the person who broke it pays. Or you split it evenly however you bought it.

Grown-ups argue about this in courts. None of the three is obviously right and all three are used by real people, so the one that matters is the one you agreed to in week 22.

If your agreement did not cover it, that is genuinely useful information and it is what week 22 was for. Add the line now, while nobody is upset. That is the second-best time and it is a lot better than the third.

Who pays for the repair?

	Repair	Bought 3/5 and 2/5	Who broke it	Split by shares	Split evenly
a	\$18.00	yes	you	\$10.80 / \$7.20	\$9.00 each
b	\$25.00	yes	them		
c	\$40.00	yes	nobody, it wore out		
d	\$12.00	yes	you		
e	\$30.00	yes	a third person		

In row c nobody broke it. Which of the three answers is left, and does that tell you anything about the other rows?

TALK ABOUT IT

Ask somebody at home what they think should happen when a shared thing breaks. Does it matter whether it was broken carelessly or by accident?

**STUCK? TRY THIS**

Work out both splits for every row before you decide anything. The decision is easier when both numbers are already in front of you.

★ CHALLENGE ZONE

Write a rule for repairs that would be fair in all five rows. Is there one?

The Sharing Page

Everything from weeks 13 to 23, on one shared thing of your own.

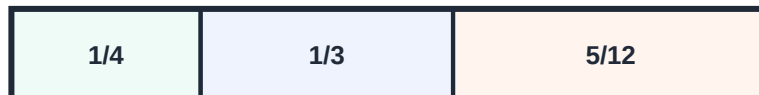
IN CLASS THIS WEEK

Word problems with unlike denominators (5.NF.2)

MONEY WARM-UP · 2 MINUTES

$$1/4 + 1/3 + 5/12 = \underline{\hspace{2cm}}$$

does it reach one whole?



the shares have to add to one whole

One page that would settle any argument

Pick something you share, or something you might. Fill in the whole picture on one page.

The cost and what each person paid. The shares as fractions, checked to reach one whole. What each share comes to in money, checked to add back to the cost. Whether you are splitting by head count or by use, and why. The five agreement lines. And anything currently owed, both columns.

If you cannot fill in a box, that is the useful part of this week. An empty box is a question nobody has asked yet, and every one of them gets harder to ask later.

The sharing page

	What it needs	Yours
a	The cost, and what each person paid	
b	The shares as fractions, adding to one whole	
c	What each share comes to, adding back to the cost	
d	By head count or by use, and why	
e	Anything owed right now, both directions	

Two of your checks should agree with each other. Which two, and what would it mean if one passed and the other failed?

► ADD TO YOUR COST BOOK

Add your finished sharing page to the Cost Book, with today's date.

TALK ABOUT IT

Show your sharing page to whoever you share the thing with.

Which box was hardest to fill in honestly?

**STUCK? TRY THIS**

Do the fraction check and the money check separately. They are the same fact twice and either one can catch a mistake the other misses.

★ CHALLENGE ZONE

Design a sharing page for three people where one of them joins halfway through. What does that change about the fractions?

The Decision With Two Numbers

Money is one of the things being compared. It is not the only one and it does not win by default.

IN CLASS THIS WEEK

Operations with decimals to hundredths (5.NBT.7)

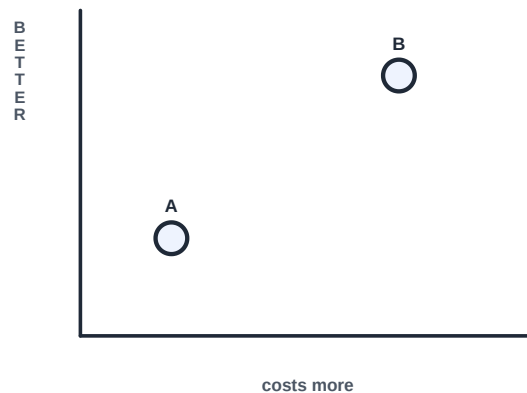
MONEY WARM-UP · 2 MINUTES

\$56.00 over 8 weeks is _____ a week

WORDS TO KEEP

two numbers

A decision where money is one of the things being compared and not the only one.



Write both, then choose

Put the money in one column and the other thing in the other. Do not add them. Do not weight them. Just get both onto the page.

What usually happens instead is that whichever number is easiest to work out becomes the decision, and money is always the easiest to work out. A thing that is twenty dollars cheaper and much worse loses to arithmetic that was never asked to consider worse.

Once both are written, say out loud which one you are choosing and what you are giving up to have it. If that sentence is hard to say, you are not finished deciding.

Two columns

	The choice	The money	The other thing	Which you would take
a	walk 20 min or pay \$2 bus	\$2.00	20 minutes	
b	\$56 now or \$7 a week for 8 weeks			
c	the good one or two cheap ones			
d	buy it or borrow a friend's			
e	fix it for \$18 or replace it for \$40			

In row b the two amounts come to the same money. So what is the decision actually about?

 TALK ABOUT IT

Take one of these home and ask somebody which they would pick. Ask why.

Is there a decision where money should not count at all?

**STUCK? TRY THIS**

Fill in the money column for every row first. Then go back and write the other thing in words rather than numbers.

★ CHALLENGE ZONE

Find a decision you made this week that had two numbers in it. What was the second one, and did you notice it at the time?

And Then a Third

Three things to weigh, and the reason people freeze.

IN CLASS THIS WEEK

Operations with decimals to hundredths (5.NBT.7)

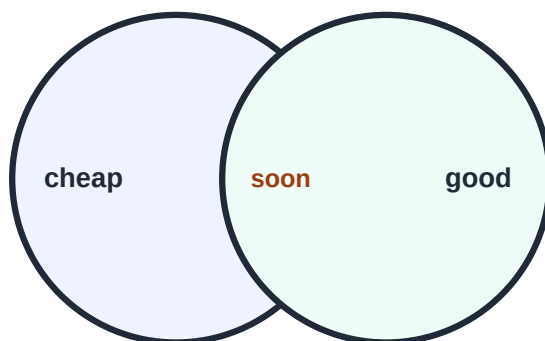
MONEY WARM-UP · 2 MINUTES

\$90 over 3 years is _____ a year

WORDS TO KEEP

two numbers

A decision where money is one of the things being compared and not the only one.



Rank them, do not score them

Money, how good it is, and how soon you get it. Three things, and no way to add them up.

There is a tempting method where you give each one a score out of ten and total them. It feels rigorous and it is not: you choose the scores, so you choose the answer, and then the arithmetic launders the choice into something that looks objective.

Do this instead. Put the three in order of what matters most to you for **this** decision. Then take the option that wins on your first one, unless it is unacceptable on another, and say which. It is quick, and it leaves a record of your reasoning that you can disagree with later.

Rank, then choose

	The decision	1st	2nd	3rd	What you would take
a	a coat for winter				
b	a snack right now				
c	a bike you will use for years				
d	a present for somebody else				
e	something for a school project due Friday				

Two of these rows put **soon** first and two put **good** first. Which are they, and what makes the difference?

 TALK ABOUT IT

Ask somebody at home to rank the three for a decision they are making.
Does your ranking change depending on how much money you have?

**STUCK? TRY THIS**

Rank before you look at any options. Ranking afterwards is just describing the choice you already made.

★ CHALLENGE ZONE

Somebody scores each option out of ten on all three and adds them up. Show how two people using that method on the same options get different answers.

What Would Change Your Mind?

Name the number that would make you decide the other way. That is most of the thinking.

IN CLASS THIS WEEK

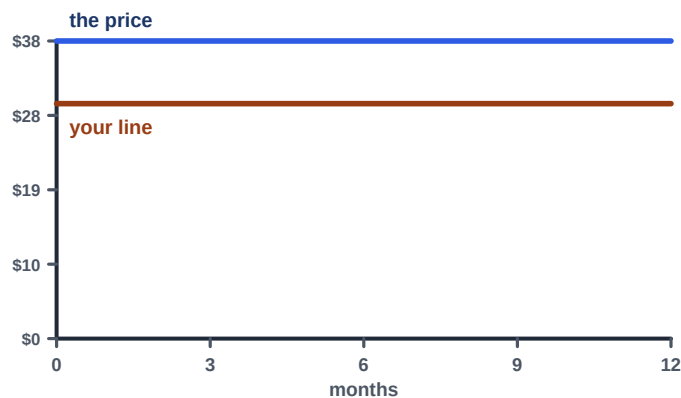
Operations with decimals to hundredths (5.NBT.7)

MONEY WARM-UP • 2 MINUTES

You would buy it at \$30. It costs \$38. The gap? _____

WORDS TO KEEP

what would change it The number that would make you decide the other way. Naming it is most of the thinking.



The most useful question in this book

You decided not to buy it. At what price would you have said yes?

If you can answer that, you did not just have a feeling, you had a reason with a number in it. If you cannot, the decision was a mood, and moods are fine but they are not repeatable and you cannot check them later.

It works in both directions and the other direction is the uncomfortable one. You are about to buy something: at what price would you have walked away? If the answer is that there isn't one, you are not deciding, you are just paying.

Name the number

	The decision	What you decided	What would change it
a	a \$38 thing you wanted	no	\$30 or less
b	a \$6 snack		
c	splitting a \$40 cost with a friend		
d	lending \$15		
e	keeping a thing you never use		

For row e the number is not a price. What kind of number is it, and what would it have to be for you to change your mind?

► ADD TO YOUR COST BOOK

Add a page headed What Would Change It, with three real decisions and the number for each.

TALK ABOUT IT

Ask somebody at home what would change their mind about something. Do not argue. Is it a weakness to have a number that would change your mind, or a strength?

**STUCK? TRY THIS**

Write the decision you actually made first. Then ask what would have to be different, and make it a number rather than a feeling.

★ CHALLENGE ZONE

Somebody says nothing would change their mind. What does that tell you about the decision? Is it ever the right answer?

Deciding for Later You

The person who lives with this decision is not in the room, and they are you.

IN CLASS THIS WEEK

Operations with decimals to hundredths (5.NBT.7)

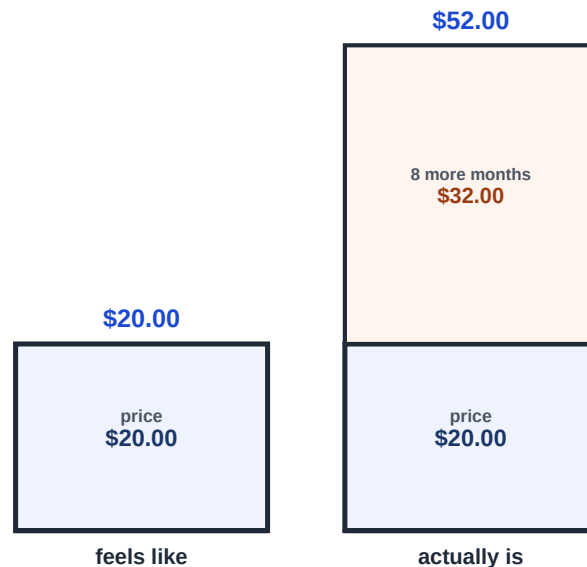
MONEY WARM-UP • 2 MINUTES

\$4.00 a month for 8 months is _____

WORDS TO KEEP

later you

The person who has to live with this decision. They are not in the room.



Today gets the thing, later pays for it

\$20.00 now and \$4.00 a month for eight months. Today you feel the twenty. The other thirty-two arrives one month at a time, at somebody who is not thinking about this decision at all.

That person is you in March. They did not get the excitement of buying it and they do get the four dollars a month, and there is no way to send them a message about it except by writing it down now.

So write it down now. In the Cost Book, with the date the last payment lands. Not as a punishment, and not to talk yourself out of anything: just so that when March arrives, the four dollars is something you chose rather than something that is happening to you.

What does later you get?

	Today	Each month	For how long	What later you pays altogether
a	\$20.00	\$4.00	8 months	\$32.00
b	\$60.00	\$1.50	2 years	
c	\$0.00	\$9.00	6 months	
d	\$140.00	\$0.00	forever	
e	\$15.00	\$2.25	18 months	

Row c costs nothing today and more than row a altogether. Which one would most people say yes to faster, and why is that the wrong instinct?

► ADD TO YOUR COST BOOK

Add a page headed For Later Me, with a cost that is still arriving and the date it stops.

TALK ABOUT IT

Tell somebody at home about a cost that is still arriving from an old decision.

Do you owe later you anything? What would they ask you for?

**STUCK? TRY THIS**

Multiply the monthly amount by the number of months. That total is the part today does not feel.

★ CHALLENGE ZONE

Write the note you would leave for yourself in March about one of these. What does it need to say to be useful rather than annoying?

Every Month, Forever

You pay, and you keep paying, and at the end you do not own it.

IN CLASS THIS WEEK

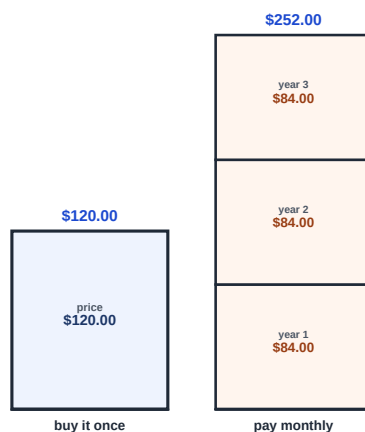
Multiply and divide multi-digit numbers (5.NBT.5, 5.NBT.7)

MONEY WARM-UP · 2 MINUTES

\$7.00 a month for 3 years is _____

WORDS TO KEEP

every month, forever A cost with no end date. You never finish paying and you never own it.



Two hundred and fifty-two dollars, and you own nothing

\$7.00 a month is \$84.00 a year and \$252.00 over three years. The same thing to buy costs \$120.00 once.

Paying monthly is not a trick. It is genuinely better when you need the thing for a short time, when you cannot put \$120.00 together at once, or when it gets replaced with a newer one while you have it. All three are real and the first two are why the arrangement exists.

What makes it worth watching is that it has no end. A price is paid and finished; a monthly cost keeps going until somebody stops it, and nobody ever schedules the stopping. Write down the month you will look at it again, or you will not look at it again.

Buy it or pay monthly

	Buy once	Or per month	1 year	3 years	When paying monthly wins
a	\$120.00	\$7.00	\$84.00	\$252.00	under 18 months
b	\$60.00	\$4.00			
c	\$200.00	\$5.00			
d	\$45.00	\$9.00			
e	\$300.00	\$6.00			

In one row you would have to keep paying for more than four years before buying it would have been cheaper. Which, and would you take that deal?

 TALK ABOUT IT

Ask somebody at home about something they pay for every month. Do not ask the amount. Why would a shop rather charge you \$7 a month than \$120 once?

**STUCK? TRY THIS**

Divide the buying price by the monthly amount. That is how many months until the monthly arrangement has cost more.

★ CHALLENGE ZONE

A monthly arrangement costs less than buying for the first two years and you expect to need the thing for three. What else would you need to know before deciding?

Own It or Borrow It

Borrowing has a price too, and for a short enough job it is the cheaper one.

IN CLASS THIS WEEK

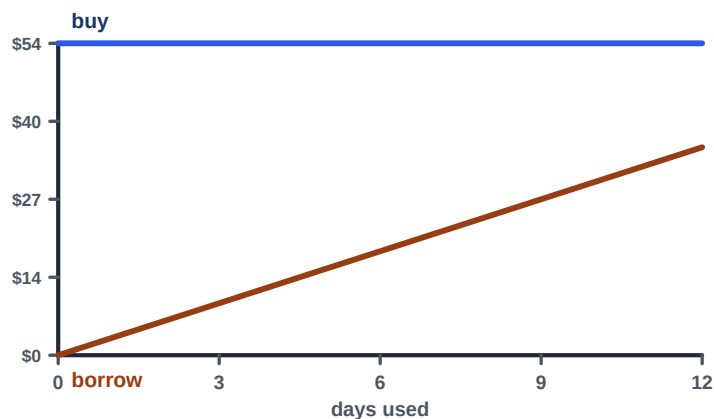
Divide with decimals (5.NBT.7)

MONEY WARM-UP · 2 MINUTES

\$3.00 a day to borrow, or \$54.00 to buy. Break even? _____

WORDS TO KEEP

own it or borrow it Two ways to have a thing, and the arithmetic that says which is cheaper for how long you need it.



Eighteen days

\$54.00 to buy, or \$3.00 a day to borrow. After eighteen days of use the borrowing has cost as much as owning, and after that it keeps going.

So the question is not which is cheaper. It is how many days you are going to use it, and that is the number people get wrong, in the direction of thinking they will use things more than they do. Week 6 already showed you your own version of that.

Borrowing also has costs with no dollar amount, which is week 10 arriving again. It has to be fetched and returned, it is not there when you want it at nine at night, and if you break it you are having a conversation about somebody else's property.

How many days until owning wins?

	Buy it for	Borrow for, per day	Days to break even	You need it for
a	\$54.00	\$3.00	18	5 days
b	\$40.00	\$8.00		3 days
c	\$90.00	\$2.50		50 days
d	\$12.00	\$5.00		2 days
e	\$150.00	\$6.00		20 days

Using the days in the last column, which rows should be bought and which borrowed? One of them is close enough that the reasons from week 10 would decide it. Which?

 TALK ABOUT IT

Find something at home that gets used a few days a year. Should it be owned?

Is there anything you would rather borrow even if owning were cheaper?

**STUCK? TRY THIS**

Divide the buying price by the daily price to get the break-even days. Then compare that with how long you actually need it.

★ CHALLENGE ZONE

A tool costs \$80 to buy and \$10 a day to borrow, and you need it for eight days this year and probably eight days next year. What do you do?

The Real Comparison

Two things you might actually buy, costed all the way through.

IN CLASS THIS WEEK

Operations with decimals to hundredths (5.NBT.7)

MONEY WARM-UP · 2 MINUTES

$$\$35 + 12 \times \$2.50 = \underline{\hspace{2cm}}$$



Everything at once, on your own decision

Pick two real options for something you might buy. Then use every tool in this book on them.

The price. The running cost, over a stretch of time you decide before you start. How long each one lasts, and what that does to the cost per year. The cost per use at your honest guess of how often, halved. Where the two options cross. And the reason with no dollar amount.

You may find they come out close, which is a real result and not a failure. Two options that are within a few dollars over three years are genuinely a coin toss on money, and that frees you to decide on the thing that is not money.

The full comparison

		Option one	Option two
a	Price		
b	Running cost over the period		
c	Whole cost over the period		
d	Cost per use, at half your guess		
e	The reason that is not money		

Write the period you chose at the top before you fill anything in. Then say which option wins on money, and whether that settles it.

TALK ABOUT IT

Show your comparison to somebody at home and ask which they would take.

Did the arithmetic change your mind, or confirm what you already wanted?

**STUCK? TRY THIS**

Choose the stretch of time first and write it down. Every row after that has to use the same one.

★ CHALLENGE ZONE

Do the whole comparison again with a period twice as long. Does the winner change? If it does, which period is the honest one?

A Decision You Already Made

Go back and cost it honestly, which is the only way this year turns into a habit.

IN CLASS THIS WEEK

Operations with decimals to hundredths (5.NBT.7)

MONEY WARM-UP · 2 MINUTES

Paid \$48, used it 6 times. Per use? _____



Not whether it was good. What it cost.

Pick something you bought a while ago. Work out what it has actually cost you, and how many times you have actually used it.

Two rules make this useful rather than miserable. Count real uses, not the ones you meant to have. And include what it has cost to keep since, which is the number the original decision definitely ignored.

You are not marking yourself. Some of these will come out well and it is worth knowing which, because the point of doing this is to find out what kind of buying decision you are actually good at. Most people are good at some and bad at others, and almost nobody knows which is which.

Cost it honestly

		What you would write
a	What did it cost to buy?	
b	What has it cost to keep since?	
c	How many times have you really used it?	
d	So what has it cost per use?	
e	What did you expect the answer to be?	

Row e is the one that teaches you something. If your expectation was a long way off, in which direction was it wrong?

► ADD TO YOUR COST BOOK

Add a page headed What It Really Cost, with one old decision costed all the way through.

TALK ABOUT IT

Do this for one thing somebody else in your house bought. Ask first.

Would you buy it again? Is that the same question as whether it was worth it?

**STUCK? TRY THIS**

Do row c before row d, and do not adjust it once you have seen what row d comes to.

★ CHALLENGE ZONE

Do this for two things, one that came out well and one that did not. What is different about the two decisions?

Was It Worth It?

A question only you can answer, using a number you worked out rather than a feeling.

IN CLASS THIS WEEK

Operations with decimals to hundredths (5.NBT.7)

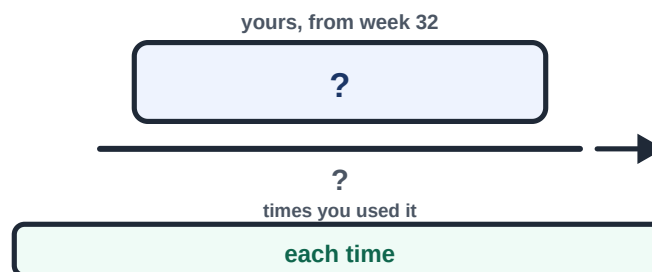
MONEY WARM-UP · 2 MINUTES

One thing cost \$0.30 a use, another \$24.00. Which was worth more? _____

WORDS TO KEEP

worth it

Something only you can decide, using a number you worked out.



The number does not decide, it informs

Thirty cents a use and twenty-four dollars a use. Those are facts. Whether either was worth it is not a fact.

A thing you used twice can still have been worth it, if those two times mattered enough. A thing you used two hundred times can have been a waste, if you never enjoyed any of them. Cost per use tells you what you paid for each go. It cannot tell you what each go was worth to you, and no arithmetic can.

What it does do is stop you being fooled. Nobody who has worked out twenty-four dollars a use still believes the thing was cheap, and nobody who has worked out thirty cents still believes the good ones were an extravagance. It clears away the wrong argument so you can have the right one.

Worth it, or not

	The thing	Cost per use	Worth it?	Why
a	used it 200 times	\$0.30		
b	used it twice, for something that mattered	\$24.00		
c	used it once, for fun	\$9.00		
d	use it every day and dislike it	\$0.05		
e	have never used it	no answer		

Row e has no cost per use at all, because you cannot divide by nothing. Say what that means, and what you would do about the thing.

 TALK ABOUT IT

Ask somebody at home about the best thing they ever bought. Was it cheap?
Can something be worth it and still have been a bad decision?

**STUCK? TRY THIS**

Fill in the third column before the fourth. The reason is the part worth writing.

★ CHALLENGE ZONE

Two people work out the same cost per use for the same thing and disagree about whether it was worth it. Who is wrong?

The One You Would Do Again

Looking for the pattern in your own decisions, which is the only reason to have kept the book.

IN CLASS THIS WEEK

Operations with decimals to hundredths (5.NBT.7)

MONEY WARM-UP • 2 MINUTES

Three things cost \$0.20, \$1.50 and \$18.00 a use. Best? _____

WORDS TO KEEP

worth it

Something only you can decide, using a number you worked out.

W
O
R
T
H

I
T

the good ones



the others



cost more

What kind of buyer are you?

Go through your Cost Book and sort your decisions into the ones you would make again and the ones you would not.

Then look for what the good ones have in common. People usually find something specific and slightly surprising. It might be that everything you waited two weeks for turned out well. It might be that everything bought with somebody else went badly. It might be that the expensive ones are all in the good pile.

Whatever it is, it is worth more than any rule in this book, because it is about you and it came out of your own numbers. Write it down as a sentence you could actually use in a shop.

Find your pattern

		Yours
a	Two decisions you would make again	
b	Two you would not	
c	What the good ones have in common	
d	What the others have in common	
e	The sentence you would say to yourself in a shop	

Row e should be short enough to remember and specific enough to act on. If it is just 'think about it', it is not finished.

TALK ABOUT IT

Say your sentence out loud to somebody at home. Ask if it sounds like you.

Is a rule about yourself more useful than a rule about money?

**STUCK? TRY THIS**

Do rows a and b from the book rather than from memory. The pattern is easier to see in four real examples than in everything you have ever bought.

★ CHALLENGE ZONE

Somebody else reads your sentence and it does not work for them. Why would a true rule about buying not transfer between two people?

Your Cost Book, Finished

Every page of it about your own money, your own things, and your own agreements.

IN CLASS THIS WEEK

Operations with decimals to hundredths (5.NBT.7)

MONEY WARM-UP • 2 MINUTES

Twelve pages at 3 decisions each is _____ decisions

one box per page of your Cost Book

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	

What is in it

Go through the Cost Book from the front. It should be in order and every page should have your own numbers on it.

A thing you own and what it has cost since. Two things costed over their first year. Three things and their cost per use. Your written guesses about how long things last. Two options and the month they cross. A shared cost with the shares checked twice. An agreement with five lines and two names. Something you owe or are owed, both columns. A cost still arriving from an old decision. And an old decision, costed honestly. Nobody else could have written it. That is the point, and it is why it is worth keeping rather than handing in.

Check the book

	Question	Your answer
a	How many pages are in your Cost Book?	
b	Which page was hardest to fill in honestly?	
c	Which number in it surprised you most?	
d	Which page would you show somebody first?	
e	Which page will still be true in a year?	

If a page is missing, fill it in now from the workbook page it came from. A book with a gap is still a book; one abandoned in March is not.

► ADD TO YOUR COST BOOK

This is the last page. Number every page in the book, front to back, and write the total on this one.

TALK ABOUT IT

Show the whole Cost Book to somebody at home, front to back.

Which page would you keep if you could only keep one?

**STUCK? TRY THIS**

Work from the front. Every page has a heading that came from a week, so any gap has a workbook page waiting to fill it.

★ CHALLENGE ZONE

If you kept this book for another year, what would you add that is not in it?

The Price Is Not the Cost

Six words, and what they are worth once you have spent a year on them.

IN CLASS THIS WEEK

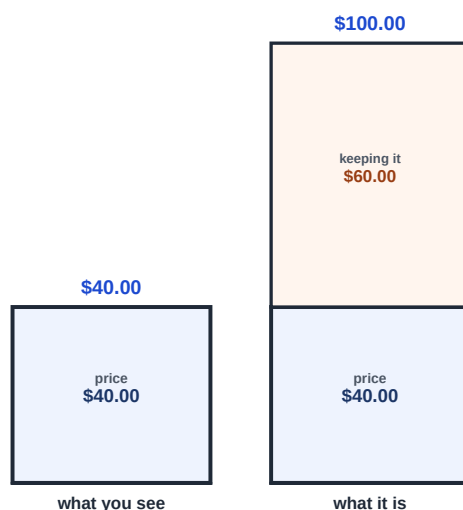
Operations with decimals to hundredths (5.NBT.7)

MONEY WARM-UP · 2 MINUTES

A \$40 thing that cost \$60 to keep. The whole cost? _____

WORDS TO KEEP

the price is not the cost The whole year in six words.



Where it goes from here

The price is the number they show you. The cost is the number you work out.

Grade 3 said every choice costs you the thing you did not pick. Grade 4 said decide it before you are standing in front of it. This year said the number on the tag is the beginning of the arithmetic rather than the end.

None of it decides for you, and that was deliberate. It makes sure that when you decide, you are deciding about the real numbers rather than the visible ones.

The last questions

	Question	Your answer
a	What is the one thing you would tell a younger student?	
b	Which idea from this year have you actually used?	
c	Which one sounded useful and you have never used?	
d	What do you now work out that you used to guess?	
e	Will you keep doing this? Say honestly.	

Row e is allowed to be no. If it is, say what would have to be different, which is a more useful answer than a polite one.

 TALK ABOUT IT

Tell somebody at home the six words and what they mean.

What is one thing you now do differently with money?

**STUCK? TRY THIS**

There is no arithmetic on this page. Answer in sentences and take your time over row c.

★ CHALLENGE ZONE

Write the week this book is missing. What would it be about, and where in the year would it go?



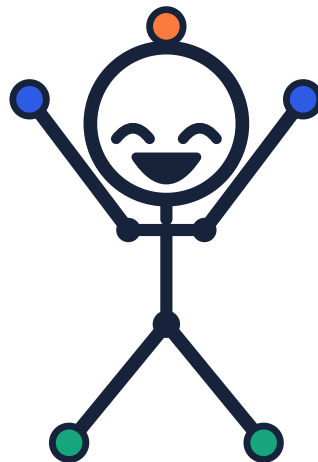
Young Innovators
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Grade 5 · The Keeping Year

Certificate of Completion

This is to certify that

completed thirty-six weeks of costing things properly: worked out what a thing costs to keep and not only to buy, divided a whole cost by the times it was used, found the month where the cheaper option stops being cheaper, split a cost with somebody else in fractions that added up, wrote an agreement down before it was needed, and learned to name the number that would change their mind.



Teacher

Date

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